Do Now · retrieval
Paper 2 examiner-style feedback

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why must recursive calls progress?

Q2

What does a midpoint calculate?

Q3

What happens when low>high?

Q4

C# [2,4,6] with a[1]=a[0]+a[2]. Final array?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Identify whether the error lies in problem reading, algorithm state or pseudocode.

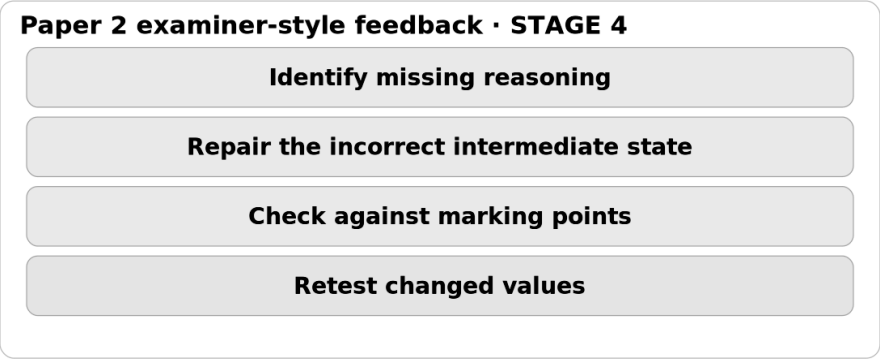
Core idea 2

Apply — Explain every bound update, recursive return or data-structure operation.

Core idea 3

Evaluate — Retest the method with a different input to demonstrate transfer.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

mark scheme — Criteria for awarding marks.

termination — Algorithm eventually stops.

auxiliary space — Additional storage beyond the input.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Repair a SumTo recursion missing a base case.

Follow the worked method

1. Explain the result — The n==0 base case stops recursion and returns zero.

Source model · final illustrated step

IDO · METHOD

The n==0 base case stops recursion and returns zero.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the result
The boundaries exclude a failed midpoint, so the remaining interval shrinks.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

An algorithm's recursion has no base case. Explain fault and give two repairs. [6]

Question 2

Trace binary search for 9 in [1,3,5,7,9], recording every low, high and midpoint. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Correct: 'merge sort has O(1) auxiliary space because the temp array is short-lived'. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Getting a correct output once guarantees an algorithm is bug-free.

Use key vocabulary from memory

1. Explain mark scheme in your own words.

2. Explain termination in your own words.

3. Explain auxiliary space in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.